Lesson 6.3 - The Geometric Distribution

> **Video:** Geometric Distribution - Probability, Mean, Variance, & Standard Deviation by The Organic Chemistry Tutor **Chapter:** 6 - Discrete Probability Distributions

The idea

A **geometric distribution** answers: *on which trial does the **first** success happen?*

If the first success is on trial x, then the first $x-1$ trials failed and trial x succeeded:

$$P(X=x)=q^{x-1}p, \qquad q=1-p$$

All the formulas

Question	Formula
First success **on** trial x	$P(X=x)=q^{x-1}p$
First success **by** trial x (within the first x)	$P(X\le x)=1-q^{x}$
First success **before** trial x	$P(X<x)=1-q^{x-1}$
First success **after** trial x	$P(X>x)=q^{x}$
First success on trial x **or later**	$P(X\ge x)=q^{x-1}$
Mean (expected trial of the first success)	$\mu=\dfrac{1}{p}$
Variance	$\sigma^2=\dfrac{1}{p}\left(\dfrac{1}{p}-1\right)=\dfrac{1-p}{p^2}$
Standard deviation	$\sigma=\sqrt{\dfrac{1-p}{p^2}}=\dfrac{\sqrt{1-p}}{p}$

Why $P(X>x)=q^x$: "no success in the first x trials" means x failures in a row. Then $P(X\le x)=1-q^x$ is its complement.

Tip for $<$ and $\ge$: shift by one so you can use the formulas above. $P(X<7)=P(X\le 6)$ and $P(X\ge 18)=P(X>17)$.

Example 1 - first six on the fourth roll

$p=\frac{1}{6}$ (rolling a 6), $q=\frac{5}{6}$:

$$P(X=4)=q^{x-1}p=\left(\tfrac{5}{6}\right)^3\cdot\tfrac{1}{6}=\frac{125}{1296}\approx 0.0965=9.6\%$$

Three failures, then one success.

Example 2 - first blue car is the 7th car

15% of cars are blue: $p=0.15$, $q=0.85$.

$$P(X=7)=q^{x-1}p=(0.85)^6(0.15)\approx 0.0566=5.7\%$$

Example 3 - the 10th person you meet is the first teacher

4% of a town's residents are teachers: $p = 0.04$, $q = 0.96$.

(a)

$$P(X = 10) = q^{x-1}\,p = (0.96)^{9}(0.04) \approx 0.0277 = 2.77\%$$

(b) Mean, variance, standard deviation

$$\mu = \frac{1}{p} = \frac{1}{0.04} = 25, \qquad \sigma^2 = \frac{1}{p}\left(\frac{1}{p} - 1\right) = 25(25-1) = 600, \qquad \sigma = \sqrt{\sigma^2} = \sqrt{600} \approx 24.49$$

Example 4 - defective tires ($p = 0.02$, $q = 0.98$)

2% of tires are defective, and a sample of tires is tested one at a time.

(a) The 8th tire is the first defective one

$$P(X = 8) = q^{x-1}\,p = (0.98)^{7}(0.02) \approx 0.0174 = 1.7\%$$

(b) The first defect is found within the first 5 tires Adding each case:

$$P(X \le 5) = p + qp + q^2p + q^3p + q^4p = p(1 + q + q^2 + q^3 + q^4) \approx 0.0961$$

With the cumulative formula, which is much faster:

$$P(X \le 5) = 1 - q^{x} = 1 - (0.98)^{5} \approx 0.0961 = 9.6\%$$

(c) The first defect is found within the first 10

$$P(X \le 10) = 1 - q^{x} = 1 - (0.98)^{10} \approx 0.1829 = 18.3\%$$

(d) How many tires do you expect to test to find the first defect?

$$\mu = \frac{1}{p} = \frac{1}{0.02} = 50 \text{ tires}$$

Example 5 - more cumulative questions (same tires)

(a) First defect before the 7th tire

$$P(X < 7) = P(X \le 6) = 1 - q^{6} = 1 - (0.98)^{6} \approx 0.1142 = 11.4\%$$

(b) First defect after the 12th tire

$$P(X > 12) = q^{x} = (0.98)^{12} \approx 0.7847 = 78.5\%$$

(c) First defect on the 18th tire or later

$$P(X \ge 18) = P(X > 17) = (0.98)^{17} \approx 0.7093 = 70.9\%$$

(d) First defect between tire 15 and tire 30 (inclusive)

Split the whole range (total probability 1) into three pieces: A = $X \le 14$, B = $15 \le X \le 30$, C = $X > 30$.

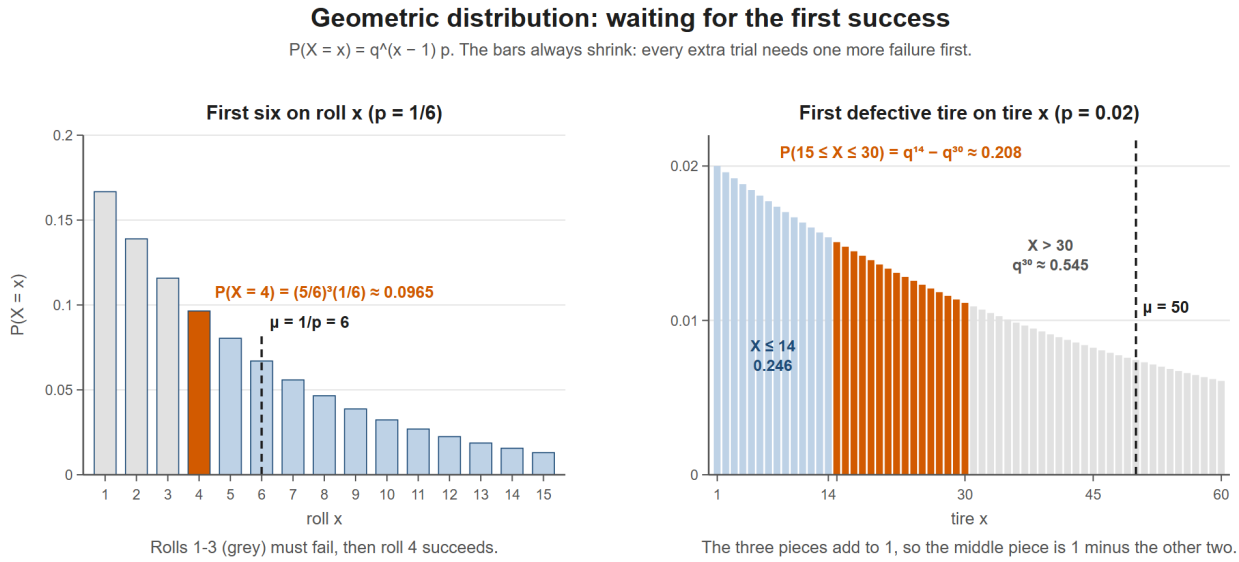
$$P(15 \le X \le 30) = 1 - P(X \le 14) - P(X > 30)$$

$$= 1 - \left[1 - (0.98)^{14}\right] - (0.98)^{30} = (0.98)^{14} - (0.98)^{30} \approx 0.7536 - 0.5455 = 0.2082 = 20.8\%$$

(e) First defect after tire 20 but no later than tire 45 ($21 \le X \le 45$)

$$= 1 - P(X \leq 20) - P(X > 45) = (0.98)^{20} - (0.98)^{45} \approx 0.6676 - 0.4029 = 0.2647 = 26.5\%$$

> **Shortcut:** $P(a \leq X \leq b) = q^{a-1} - q^{b}$. Check: (d) is $q^{14} - q^{30}$ ✓ and (e) is $q^{20} - q^{45}$ ✓. Make sure no trial is counted twice or left out.



Left: the first six on roll 4 needs three failures (grey) and then a success. Right: for the tires, the range 15 to 30 (orange) is whatever is left after removing X ≤ 14 and X > 30, which is q^14 − q^30.

Key takeaways

- Geometric = waiting for the **first** success: $P(X = x) = q^{x-1}p$.
- Cumulative: $P(X \leq x) = 1 - q^{x}$ and $P(X > x) = q^{x}$. Convert $<$ and $\geq$ by shifting by 1.
- $\mu = 1/p$ is the expected number of trials until the first success.